MILITARY SPECIFICATION

Control, Radio Set C-2790(*)/ARC; Indicator, Frequency Channel ID-758(*)/ARC; Mounting MT-2141(*)/ARC and Switching Unit SA-629(*)/ARC

This specification has been approved
by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this Specification shall provide the following:

(1) Primary control of the operating functions of the Receiver-Transmitter RT-332/ARC-52.

(2) Remote indication of the UHF communications preset channel or the manual frequency.

(3) On-Off Control of the ADF equipment.

(4) Optional mode control of the Auxiliary Receiver of the AN/ARR-40.

1.2 Classification - The equipment covered by this specification consists of the following items:

Item	Type Designation	Appl. Para.
Indicator, Frequency Channel	ID-758/ARC	3.4.1
Control, Radio Set	C-2790/ARC	3.4.2
Switching Unit	SA-629/ARC	3.4.3
Mounting	MT-2141/ARC	3.4.4

FSC-5895

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:18Z
Check the source to verify that this is the current version before use.

1.3 Associated Equipment - The equipment shall be capable of operating with the Receiver-Transmitter RT-332/ARC-52 which is not to be supplied as part of this equipment.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS:

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental testing, Aeronautical and associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-P-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, format for

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:18Z
Check the source to verify that this is the current version before use.

MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-E-19600	Electronic Modules, General Specification for
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-L-25467	Lighting; Integral, Instrument

STANDARDS:

Military

MS 25212	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions
MS 25213	Control Panel, Aircraft Equipment, Typical Installation
MS 25245	Connections Input Power Airborne Electronic Equipment; 28-volt dc, 115-volt Single-Phase ac, and/or 115/200 volt Three-Phase ac, or Combinations
MS 33556	Case, Instrument, 3-1/4" size Standard Dimensions for
MS 33558	Numerals and Letters; Aircraft
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

PUBLICATIONS

Bureau of Naval Weapons

SAR-300	Engineering Changes and Material Bulletins, Preparation of

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the currentversion before use.

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by the contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Ave., Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Contruction

3.2.1 General - This equipment shall conform with all applicable design, workmanship, and construction requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Switches - Rotary switches mechanically positioned by other than manual means may be mounted by standoffs.

3.2.3 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 12.2 lbs.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:18Z
Check the source to verify that this is the current version before use.

3.2.4 Operating Life

3.2.4.1 Operational Stability - The equipment shall operate with optimum performance for 20,000 cycles without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 100,000 cycles with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5 Cabling and Connections

3.2.5.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specifications MIL-E-5400.

3.2.5.2 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of connectors attached to the cables shall not be supplied as part of the equipment.

3.2.5.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.6 Control Panels - All console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The confirguration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.7 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.8 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.9 Maintenance Provisions and Field Testing - Provisions for maintenance shall be specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:18Z
Check the source to verify that this is the currentversion before use.

3.2.10 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.11 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests.

Temperature	Room Ambient (30°C ±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Power Voltage	27.5 ±0.5 V DC

3.2.12 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment, with the following exceptions:

3.2.12.1 Altitude - The equipment shall operate satisfactorily at an altitude of 70,000 feet.

3.2.12.2 Vibration - The Indicator, Frequency Channel ID-758 shall operate satisfactorily when subjected to the vibration environment as specified in procedure IV and V of Specification MIL-E-5272. The double amplitude shall be .018 - .020 inch and all tests shall be conducted in each of three mutually perpendicular planes in lieu of the circular vibration specified. The vibration requirements for the control unit, switching unit and mounting base shall be as specified in Specification MIL-E-5400, curves I and II.

3.2.13 Warmup Time - The warmup time for the equipment shall not exceed 1 minute under standard conditions and 2 minutes at extreme service conditions.

3.2.14 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC Power 28V Category "B"

Control, Radio Set	0.15 amp
Switching Unit	1.75 amps
Indicator, Frequency Channel	0.75 amp

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:18Z
Check the source to verify that this is the current version before use.

3.2.14.1 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.14.2 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.15 Fail-Safe Provisions - Adequate fail-safe provisions shall be provided to permit rapid selection of emergency transmission and reception on the guard frequency. Circuitry shall not be interconnected with the memory drum.

3.3 Performance

3.3.1 Control Functions - The Control, Radio Set C-2790/ARC and the Switching Unit SA-629/ARC shall provide the means for controlling all operating functions of the Receiver-Transmitter RT-332/ARC-52 and the Indicator, Frequency Channel ID-758/ARC.

3.3.2 Visual Displays - The Indicator, Frequency Channel ID-758/ARC shall provide display of the following:

(1) Manual Frequency

(2) Preset Channel

(3) Guard

(4) Off

Simultaneous display of the above information will not be required.

3.4 Detail Requirements

3.4.1 Indicator, Frequency Channel - ID-758/ARC

3.4.1.1 Function - The Indicator shall provide remote indication of the UHF communications preset channel or the manual frequency as determined by the Control, Radio Set C-2790/ARC.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

3.4.1.2 Form Factor - The indicator shall conform to the type of mounting and general form factor of MS 33556. The maximum dimensions shall not exceed 3¼ by 1 5/16 by 6 11/16 inches.

3.4.1.3 Weight - The total weight of the indicator shall not exceed 1.3 pounds.

3.4.1.4 Contents - The indicator shall contain the necessary automatic positioning devices, indicator means, integral lighting and all other parts necessary to make the equipment function as a unit.

3.4.1.5 Controls, Internal - There shall be no internal, adjustable controls.

3.4.1.6 Electrical Connection - All connections to the indicator shall be made through one connector, mounted on the rear of the indicator.

3.4.1.7 Integral Lighting - The indicator shall be integrally lighted in accordance with MIL-L-25467 except that 28 volt lamps shall be used.

3.4.1.8 Power Shutter Mechanism - A shutter mechanism shall be provided for partially obscuring the indicator display. The shutter mechanism shall be excited by 27.5 v dc and excitation shall cause the shutter to disappear from view. The 27.5 v d-c excitation ground shall be obtained from the main power switch on the Control, Radio Set C-2790/ARC.

3.4.1.8.1 Operating Current - The shutter mechanism operating current shall be 130 milliamperes maximum at 27.5 v dc under standard conditions.

3.4.1.8.2 Lettering - The word "OFF" shall be on the shutter mask in black letter approximately 3/32 inches high on a red background color no. 66 of Specification MIL-D-25177. The letter style shall be in accordance with MS 33558.

3.4.1.9 Guard Shutter Mechanism - A shutter mechanism shall be provided for obscuring the indicator frequency or channel display and exposing the indication of Guard channel type. The shutter mechanism shall be excited by 27.5 v dc and voltage excitation shall cause the shutter display to be exposed in the viewing window. The excitation voltage shall be obtained from the function selector switch in the Control, Radio Set C-2790/ARC.

3.4.1.9.2 Lettering - The word "GUARD" shall be on the shutter mask in white letters approximately 5/32 inch high on a dull black background. The letter style shall be in accordance with MS 33558.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

3.4.1.10 Channel/Frequency Shutter - A shutter mechanism shall be provided for exposing either the frequency or channel display. The shutter mechanism shall be excited by 27.5 v dc. In the deenergized position the 2 left-hand digits on the indicator display shall be exposed for indication of the preset channel selection. In the energized position, all 4 digits on the indicator shall be exposed. The excitiation voltage shall be obtained from the function slector switch in the Control, Radio Set C-2790/ARC.

3.4.1.10.1 Operating Current - The shutter mechanism operating current shall be 130 milliamperes maximum at 27.5 v dc under standard conditions.

3.4.1.10.2 Lettering - Markings on the shutter mask shall not be required. Indication of channel or frequency information shall be determined by the number of digits exposed in the viewing window. 2 digits exposed shall indicate the UHF channel and 4 digits exposed shall indicate the UHF Manual Frequency.

3.4.1.11 Indications - The indicator shall be capable of displaying the following information:

(1) UHF preset channel - indication "1" through "20"
or
UHF manual frequency - indication "225.0" through "399.9"

(2) Guard channel - indication "GUARD"

3.4.1.12 Sequence of Display - The indicator shall be marked such that the numerals in the window travel upward for increasing numerical values of channel, or frequency. The indicator shall provide bi-directional operation. Each number shall be driven by an intermittent motion type mechanism rather than a continuous type drive.

3.4.1.13 Duty Cycle - The maximum operating time of the indicator shall not exceed 2 seconds on and 4 seconds off.

3.4.1.14 Numerals - All display numerals shall be in accordance with Military Standards MS 33558 and approximately 1/4 inch high.

3.4.1.15 Visibility of Indications - With the shutter mechanism deflected the indications shall be legible from any point within a truncated polyhedron, the smaller end of which is the instrument aperture. The top, bottom and sides of the polyhedron are inclined at an angle of 45°, 15°, and 30° respectively with a line perpendicular to the indicator mounting surface.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

3.4.1.16 Display Interaction - When a selection or tuning process is being performed, the visual display and tuning information of the function not being adjusted shall be retained by the switching unit. Upon selection of either channel or frequency type operation, the receiver-transmitter and the remote indicator shall immediately resume the last-made selection.

3.4.1.17 Hermetic Sealing - The Indicator, Frequency Channel ID-758/ARC shall be hermetically sealed. The case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures (soldering irons are considered to be standard tools). The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the component may be subjected. The case shall be such that it will meet the pressure requirements necessary when tested for leaks by means of a mass spectrometer type of helium leak detector.

3.4.1.17.1 Filling Medium - The Filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.4.1.17.2 Leak Rate - The indicator shall be sealed such that the initial maximum detected leak rate at a pressure differential of approximately one atmosphere shall not exceed 1.0 micron cubic foot per hour.

3.4.2 Control, Radio Set C-2790/ARC - The control shall be of the console type in accordance with Specification MIL-C-6781 and shall fulfill the following requirements.

3.4.2.1 Function - The control shall provide primary control over all operating functions of the Receiver-Transmitter RT-332/ARC-52 and the Indicator, Frequency Channel ID-758/ARC.

3.4.2.2 Form Factor and Mounting - The control shall conform to the basic form factor of MS 25212. The mounting and installation shall conform to MS 25213. The dimension parallel to the mounting rails shall not exceed 3.0 inches.

3.4.2.3 Controls, Internal - There shall be no internal adjustable controls.

3.4.2.4 Electrical Connection - All connections shall be made through one connector, mounted on the rear of the control.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

3.4.2.5 Illumination - Red lighting and a plastic light plate in accordance with Specification MIL-P-7788 shall be provided.

3.4.2.6 Controls - The following controls shall be provided on the face of the control:

3.4.2.6.1 Volume-Power Control - A rotary thumb wheel type volume control shall be provided to adjust the amplitude of the audio signals delivered to the headset circuits. A suitable switch shall be mechanically connected to the main control shaft to control the main power to the Receiver-Transmitter RT-332/ARC-52 and the Indicator, Frequency Channel ID-758/ARC. When in the extreme counterclockwise position the main power shall be disabled. The plastic lighting plate shall be marked "COMM VOL" and the word "OFF" shall be suitably marked on the thumb wheel surface.

3.4.2.6.2 Mode of Operation Control - A three position rotary switch having optional identification and mode of operation selection shall be provided. The procuring activity shall determine and specify the type to be used for each installation. Both units are internally identical. The addition of the Auxiliary Receiver (AN/ARR-40) function requires different external connections.

For use with the RT-332/ARC-52, the following identification and mode of operation selection shall be provided:

(1) Panel marked "T/R" - permits selection of transmission and reception on the UHF main frequency.

(2) Panel marked "T/R+G" - permits selection of transmission and reception on the UHF main frequency and reception only on the UHF guard frequency.

(3) Panel marked "ADF" - permits selection of Automatic Direction Finding operation on the UHF main frequency.

For use with Integrated Electronic Centrals AN/ASQ-19, AN/ASQ-56, and AN/ASQ-57, the following identification and mode of operation selection shall be provided:

(1) Panel marked $\overset{\text{"T/R"}}{\text{ADF}}$ - permits selection of transmission and reception on the UHF main frequency and selection of Automatic Direction Finding operation on the Auxiliary Receiver frequency.

(2) Panel marked $\overset{\text{"T/R+G"}}{\text{ADF}}$ - permits selection of transmission and reception on the UHF main frequency, reception only on the UHF guard frequency, and selection of Automatic Direction Finding operation on the Auxiliary Receiver frequency.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

(3) Panel marked "ADF" G - permits selection of Automatic Directing Finding operation on the UHF main frequency and reception only on the Auxiliary Receiver guard frequency.

3.4.2.6.3 Function Selector Control - A three-position toggle switch having the following identification and function selection shall be provided:

(1) Panel marked "FREQ" - permits selection of manual frequency operation of the Receiver-Transmitter RT-332/ARC-52.

(2) Panel marked "GUARD XMIT" - permits selection of reception and transmission on the emergency UHF communications guard frequency.

(3) Panel marked "CHAN" - permits selection of channel type operation of the Receiver-Transmitter RT-332/ARC-52.

3.4.2.6.4 Frequency/Channel Controls - Three rocker type actuators shall be provided to permit changing of the UHF communications preset channel or frequency. When the function selector control, 3.4.2.6.3, is in the "CHAN" position, the left-hand actuator shall be capable of changing the preset channel. The center and right-hand actuators shall be disabled electrically. When the function selector control is in the "GUARD XMIT" position, all three actuators shall be disabled electrically and the switching unit shall tune the Receiver-Transmitter RT-332/ARC-52 to the guard frequency. When the function selector control is in the "FREQ" position the left-hand actuator shall be capable of changing the 10 mc manual digit, the center actuator the 1.0 mc digit and the right-hand actuator the 0.1 mc digit.

3.4.2.7 Weight - The total weight of the control shall not exceed 1.7 pounds.

3.4.3 Switching Unit SA-629/ARC

3.4.3.1 Function - The switching unit shall provide the necessary interconnections between the Control, Radio Set C-2790/ARC, the Indicator, Frequency Channel ID-758/ARC, and the Receiver-Transmitter RT-332/ARC-52 to establish a complete function.

3.4.3.2 Form Factor - The switching unit shall be 3-9/16 inches wide by 5-13/16 inches high by 10-1/8 inches long, including the dust cover. These dimensions do not include projection of rear connector, handles, special access provisions or insertion-extraction mechanism beyond the front panel.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

3.4.3.3 Contents - The switching unit shall contain the necessary automatic tuning devices, frequency selection system, and other parts to make the equipment function as a unit and provide the necessary information for the Indicator, Frequency Channel ID-758/ARC and the Receiver-Transmitter RT-332/ARC-52. Provisions shall be incorporated to permit operation of a dual Indicator, Frequency Channel ID-758/ARC.

3.4.3.4 Controls, Internal - The following controls shall be mounted internally but shall be accessible as specified.

3.4.3.4.1 Memory Drum - A memory drum shall be provided for storage of up to 20 present channels. This drum shall be accessible from the front of the switching unit by means of a hinged door on the front panel and it shall not be necessary to remove the switching unit from the mounting.

3.4.3.4.2 Guard Channel - A terminal board shall be provided for presetting the guard frequency. It shall be possible to preset the guard frequency within the range of 225.0 mc to 399.9 mc. Unless otherwise specified, the guard frequency shall be set at 243.0 mc. Access for changing the guard channel frequency may be obtained by removing the switching unit from its mounting, removing the dust cover, and the wrap around front end cover. Proper instructions shall be given on the terminal board to permit changing the guard frequency by buss wiring revision.

The guard channel terminal board shall be self-contained and not integral with the memory durm.

3.4.3.5 Electrical Connection - All connections to the switching unit shall be made through one connector. A sutiable insertion-extraction mechanism shall be provided to permit convenient connection to the Mounting MT 2141/ARC.

3.4.3.6 Weight - The total weight of the switching unit shall not exceed 8 pounds.

3.4.3.7 Modular Construction - The switching unit shall employ modular type construction in accordance with Specification MIL-E-19600.

3.4.3.7.1 Mounting Dimensions and Hardware - The location of the nominal centerline of all hold-down screws in a given module shall be as shown in Fig. 1. The modules shall be mounted with a fastening device terminating in an 8-32NC-2A thread.

Source: https://assist.dla.mil -- Downloaded: 202610-19T13:18Z
Check the source to verify that this is the current version before use.

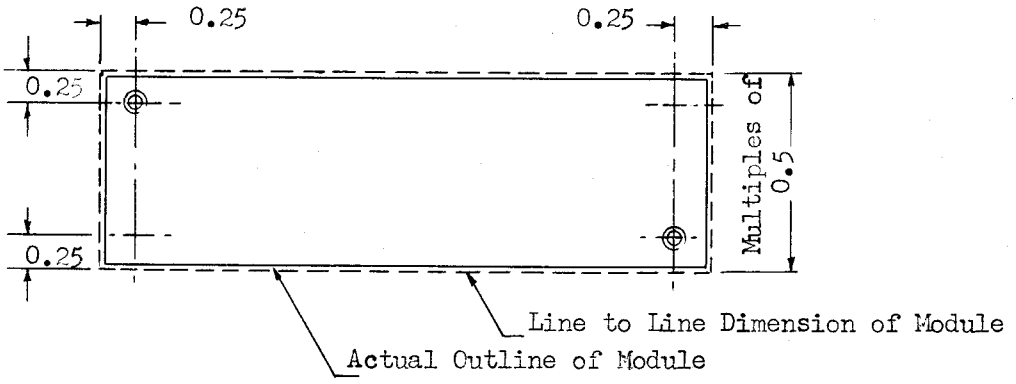


Figure 1 - Mounting Dimensions

3.4.4 Mounting MT-2141/ARC

3.4.4.1 Function - The mounting shall provide the necessary mounting provisions for the Switching Unit SA-629/ARC.

3.4.4.2 Form Factor - The mounting shall be 5-29/32 inches wide by 13-23/32 inches long by 7-9/16 inches high including sway space, front end hold-down or latching mechanism and electrical connector.

3.4.4.3 Electrical Connection - All connections to the mounting shall be made through one connector attached to the rear of the mounting.

3.4.4.4 Weight - The total weight of the mounting shall not exceed 1.2 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine conformance with all the applicable requirements.

(1) Preproduction Test
(2) Production Sample Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipments to the specified destination for service approval tests. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all requirements of this specification and the contract. Preproduction tests shall include testing in accordance with Specifications MIL-T-5422 and MIL-I-6181, except that vibration testing for the Indicator, Frequency Channel ID-758/ARC shall be in accordance with 3.2.12.2.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:18Z
Check the source to verify that this is the current version before use.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction and performance and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling

(3) Special Tests

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Tests

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated for 100 operations in both Channel and Frequency without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 55 cps (excluding resonant points). A minimum double amplitude of .020 inches shall be used for the Remote Indicator ID-758/ARC and a minimum amplitude of ±2 g's shall be used for the Radio Set Control C-2790/ARC and Switching Unit SA-629/ARC.
Duty Cycle	Not to exceed that specified for each unit.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the 100 operations. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the operating test following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests:

Quantity of Equipments Delivered for Acceptance	Quantity To Be Tested
First 10	1
Next 50	1
Next 75	1
Next 100	1
1 for each additional 200 or fraction thereof	

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run

(3) On an early equipment after an engineering change

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring agency. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such test required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests (where applicable)

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect

(3) Submit to the procuring activity proposed corrective action intended to reduce the possiblity of the same failure(s) occurring in future tests

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

4.5 Life Test - A 50,000 operation life test shall be conducted at the contractor's plant on an early equipment(s) representative of the current production. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following: When a production sample equipment is submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life test.)

Quantity of Equipments Offered for Acceptance	Quantity To Be Tested
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal ground (0-5000 ft)
Humidity	Room ambient
D.C. Voltage	26.5 ± 2.5 volts
Duty Cycle	Not to exceed that specified for each equipment

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 2,500-cycle intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the current version before use.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of operations which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the procuring activity will furnish the procedures to be used for testing.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

4.7 Presubmission Testing - No item, part of complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, will all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejections and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.9 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. ECP's, MC's and MB's shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 $\pm$ 0.5 dc. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The Contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 20260-19T13:18Z
Check the source to verify that this is the current version before use.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified herein.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400 it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not be be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use of sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T13:18Z
Check the source to verify that this is the currentversion before use.